## Abstract

In this experiment, we observed how a p-n junction silicon diode behaves when it is connected in a forward-bias circuit. We built a series loop on a breadboard using an adjustable DC source, a 1k ohm resistor, and the diode. By slowly turning up the input voltage, we monitored the changing voltage drops across the components and calculated the circuit current. The collected data was then used to plot the I-V characteristic curve of the diode. From this graph, we successfully identified the knee voltage and calculated both the static and dynamic resistance values of the device.

## Introduction

**Objectives:**

- To become familiar with a semiconductor diode and identify its terminals.
- To determine the current-voltage (I-V) characteristic curve of the diode.
- To find out the different parameter values of the diode from the experimental data.

**Motivation:**

The main motivation behind this experiment is to see how a p-n junction diode works as a basic electronic switch. Understanding how a diode allows current to pass only in one direction after crossing a specific voltage barrier is essential for learning about more advanced devices. This practical work helps us understand the difference between theoretical textbook equations and the actual performance of physical hardware components.

## Theory

**Forward Bias:** When we connect the positive side of the power supply to the anode (p-side) and the negative side to the cathode (n-side), it is called forward bias. In this state, the diode starts conducting current once the supply voltage overcomes the internal barrier potential (around 0.7 V for silicon).

**Knee Voltage ($V_k$):** This is the specific voltage level where the diode current starts to increase rapidly. Based on our lab data, we found this to be 0.6020 V.

**Resistance:**

**Static Resistance ($r_s$):** It is the resistance at a specific point on the curve, calculated as

$r_s = V_d / I_d$ .

**Dynamic Resistance ($r_d$):** It is the resistance for a small change in voltage and current, calculated as $r_d$ = Delta $V_d$ / Delta $I_d$ .

## Apparatus

| SL# | Apparatus | Quantity |
|---|---|---|
| 1 | Diode | 1 |
| 2 | Resistance (1 k ohm) | 1 |
| 3 | Project Board | 1 |
| 4 | DC Power Supply | 1 |
| 5 | Multimeter | 1 |

## Methodology

The step-by-step process we followed in the lab is described below:

**Measuring Resistor:** We took a 1 k ohm resistor and checked its actual value using a digital multimeter in ohmmeter mode.

**Circuit Setup:** We placed the silicon diode and the resistor on a breadboard. We connected them in a series circuit with a DC power supply. We made sure the anode of the diode was connected towards the positive side of the source.

**Voltmeter Connection:** We used multimeters to measure the voltages. One was connected across the diode to measure $V_d$ and another was used to measure the voltage across the resistor ($V_R$) .

**Taking Readings:** We started with the supply voltage at 0 V . Then, we slowly increased the source voltage ($E_s$) in small steps of 0.1 V until we reached 1.0 V . After that, we increased the voltage in 1.0 V steps until we reached a maximum of 5.0 V.

**Data Collection:** At each step, we recorded the values of $V_d$ and $V_r$ from the meters. Later, we calculated the diode current ($I_d$) by using the formula $I_d = V_R / R$ .

## Simulation and Results

### Data Table

| **Source Voltage $E_s$ (V)** | **Diode Voltage $V_D$ (V)** | **Resistor Voltage $V_R$ (V)** | **Diode Current $I_D$ (V)** |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0.1 | 0.1425 | 0.0011 | 0.000 |
| 0.2 | 0.2613 | 0.0007 | 0.000 |
| 0.3 | 0.2898 | 0.0015 | 0.006 |
| 0.4 | 0.3932 | 0.0171 | 0.029 |
| 0.5 | 0.4476 | 0.0620 | 0.060 |
| 0.6 | 0.4890 | 0.1453 | 0.164 |
| 0.7 | 0.5160 | 0.2527 | 0.205 |
| 0.8 | 0.5212 | 0.2847 | 0.315 |
| 0.9 | 0.5400 | 0.4168 | 0.367 |
| 1.0 | 0.5445 | 0.4483 | 0.478 |
| 2.0 | 0.6020 | 1.3945 | 1.426 |
| 3.0 | 0.6283 | 2.378 | 2.420 |
| 4.0 | 0.6473 | 3.430 | 3.364 |
| 5.0 | 0.6600 | 4.396 | 4.408 |

## Calculations

The diode current ($I_d$) for each step in the data table was calculated using Ohm's Law across the resistor:

$$I_d = \frac{V_R}{R}$$

Where, the measured resistance value is $R = 1\ \text{k}\Omega = 1000\ \Omega$.

Sample Calculation (For $E_s = 5.0\ \text{V}$):

- Measured Resistor Voltage, $V_R = 4.396\ \text{V}$
- Measured Diode Voltage, $V_D = 0.660\ \text{V}$

1. Diode Current ($I_d$):

$$I_d = \frac{4.396\ \text{V}}{1000\ \Omega} = 0.004408\ \text{A} = 4.408\ \text{mA}$$

2. Static Resistance ($R_{DC}$):

$$R_{DC} = \frac{V_D}{I_d} = \frac{0.660\ \text{V}}{4.408 \times 10^{-3}\ \text{A}} \approx 149.73\ \Omega$$

3. Dynamic Resistance ($r_{ac}$) between $4.0\ \text{V}$ and $5.0\ \text{V}$:

$$\Delta V_D = 0.6600\ \text{V} - 0.6473\ \text{V} = 0.0127\ \text{V}$$

$$\Delta I_d = 4.408\ \text{mA} - 3.364\ \text{mA} = 1.044\ \text{mA} = 1.044 \times 10^{-3}\ \text{A}$$

$$r_{ac} = \frac{\Delta V_D}{\Delta I_d} = \frac{0.0127\ \text{V}}{1.044 \times 10^{-3}\ \text{A}} \approx 12.16\ \Omega$$

**Original Data Sheet**

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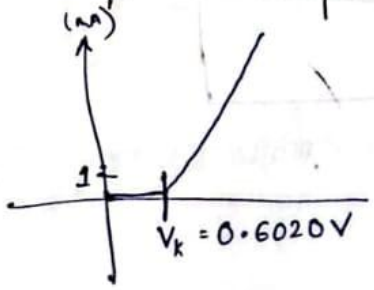
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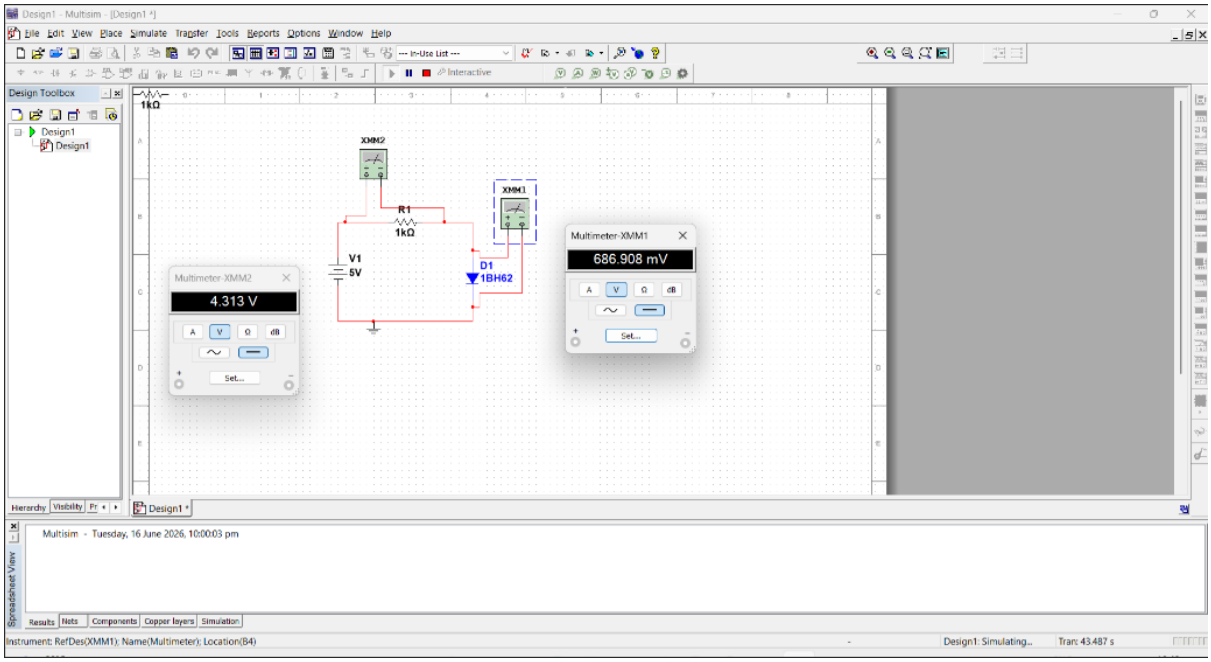
| Source voltage $V_S$(V) | Diode voltage $V_D$(V) | Resistor voltage $V_R$(V) | Diode Current $I_d$ (mA) |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0.1 | 0.1425 | 0.0001 | 0.000 |
| 0.2 | 0.2613 | 0.0007 | 0.000 |
| 0.3 | 0.2898 | 0.0015 | 0.006 |
| 0.4 | 0.3932 | 0.0171 | 0.029 |
| 0.5 | 0.4476 | 0.0602 | 0.060 |
| 0.6 | 0.4890 | 0.1453 | 0.164 |
| 0.7 | 0.5160 | 0.2527 | 0.205 |
| 0.8 | 0.5212 | 0.2847 | 0.315 |
| 0.9 | 0.5400 | 0.4168 | 0.367 |
| 1.0 | 0.5445 | 0.4483 | 0.478 |
| 2.0 | 0.6020 ($V_k$) | 1.3945 | 1.426 |
| 3.0 | 0.6283 | 2.378 | 2.420 |
| 4.0 | 0.6473 | 3.430 | 3.364 |
| 5.0 | 0.6600 | 4.396 | 4.408 |

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## Simulated Results (Simulated Diagrams)



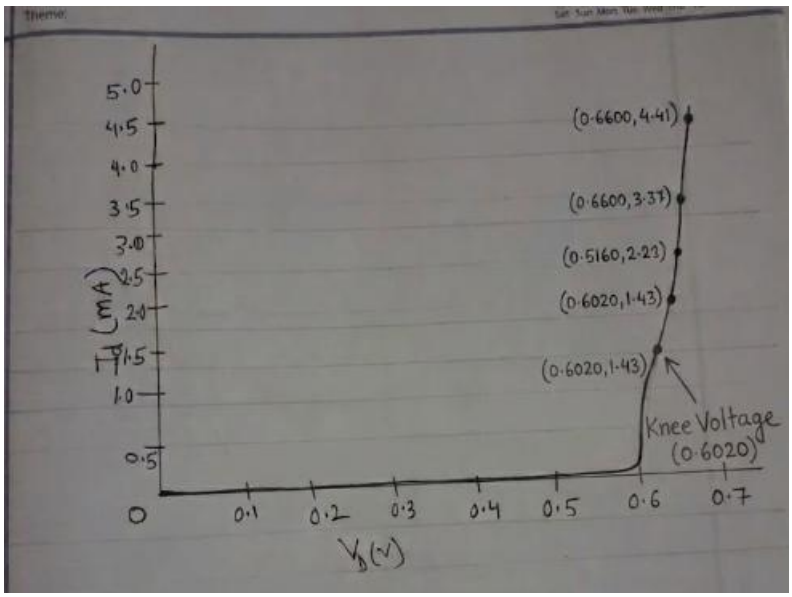

## Questions with Answers

1.Show the difference between your simulated and measured values. Comment on the results.

For $E_s = 5$ V:

- Measured values: $V_D = 0.660$ V, $V_R = 4.396$ V.
- Simulated values: $V_D = 0.687$ V, $V_R = 4.313$ V.
- **Comment:** The simulated and measured values are very close to each other. The minor differences occur due to the internal resistance of real wires and instrument errors in the laboratory.

2. Plot the $V_D$ – $I_D$ characteristic curve for the diode and comment on the graph.




Comment: The graph shows that at low voltages, the diode current stays at zero. Once the voltage crosses the knee voltage, the current increases rapidly in a non-linear way.

3. Compute the knee voltage and dynamic and static resistances of the diode from the plotted graph.

- **Knee Voltage ($V_k$):** From the data, $V_k \approx 0.602\ \text{V}$.
- **Static Resistance ($R_{DC}$):** Taken at $E_s = 5\ \text{V}$ (where $V_D = 0.660\ \text{V}$ and $I_D = 4.408\ \text{mA}$):

$$R_{DC} = \frac{V_D}{I_D} = \frac{0.660\ \text{V}}{4.408 \times 10^{-3}\ \text{A}} \approx 149.73\ \Omega$$

- **Dynamic Resistance ($r_{ac}$):** Taken between $E_s = 4\ \text{V}$ and $E_s = 5\ \text{V}$:

$$\Delta V_D = 0.6600\ \text{V} - 0.6473\ \text{V} = 0.0127\ \text{V}$$

$$\Delta I_D = 4.408\ \text{mA} - 3.364\ \text{mA} = 1.044\ \text{mA}$$

$$r_{ac} = \frac{\Delta V_D}{\Delta I_D} = \frac{0.0127\ \text{V}}{1.044 \times 10^{-3}\ \text{A}} \approx 12.16\ \Omega$$

4. What will happen if the polarity of the supply voltage is reversed in the case of using a diode with a PIV of 4.8 V?

If the supply voltage is reversed up to 5 V or higher, it will exceed the Peak Inverse Voltage (PIV) limit of 4.8 V . The diode will enter the avalanche breakdown region, conduct heavy reverse current, and get permanently damaged.

5. Discuss the overall aspects of the experiment. Did your results match the expected ones? If not, explain.

Yes, the experimental results perfectly matched the expected theoretical behavior of a silicon diode. The current remained zero until the threshold voltage was reached, after which it rose exponentially. The real-world measurements and simulation graphs confirmed this forward-bias characteristic.

## Discussion & Conclusion

Discussion:

In this experiment, we successfully observed the behavior of a silicon diode in forward-bias mode. From our recorded data we can see that when the source voltage was low (from 0 V to 0.4 V ), the diode current was almost zero or very small. This is because the voltage was not enough to cross the barrier potential. However, once the diode voltage reached around 0.6020 V, the current started increasing rapidly. This value is our Knee Voltage ($V_k$). We also noticed that even if we increased the source voltage up to 5.0 V, the voltage across the diode stayed mostly between 0.6 V and 0.66 V , which shows the typical characteristic of a silicon diode. There might be some small errors in the readings due to the internal resistance of the wires or the multimeter accuracy, but the overall curve matched the theory.

Conclusion:

The objectives of this experiment were fully fulfilled. We learned how to connect a diode in a circuit and how to measure its characteristics practically. By plotting the data, we confirmed that a silicon diode requires a specific minimum voltage to conduct current. This experiment helped us understand the fundamental working principle of semiconductor devices which will be very useful for our future electronic projects.